

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1280

Goose Haven 100 kV
January 02, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1280. The Interconnection Customer proposes a 182.6 MW solar photovoltaic facility interconnecting at the Goose Haven 100 kV substation in the Illinois area. The request was received on September 01, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 437 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,687,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: STAPLEY-ROSEMORE B 100 kV Line
Contingency: STAPLEY-ROSEMORE B 100 kV Circuit 2
Loading: 133.4% (Rating: 198 MVA)
DFAX: 33.0%
Pre-project loading: 103.0%
2. Facility: STAPLEY-ROSEMORE B 100 kV Circuit 2
Contingency: STAPLEY-ROSEMORE B 100 kV Circuit 1
Loading: 133.4% (Rating: 198 MVA)
DFAX: 33.0%
Pre-project loading: 103.0%
3. Facility: GOOSE HAVEN-ROSEMORE B 100 kV Ckt 2
Contingency: ROSEMORE B-GOOSE HAVEN SUBSTATION 100 kV Circuit 1
Loading: 128.6% (Rating: 237 MVA)
DFAX: 35.3%
Pre-project loading: 101.5%
4. Facility: STAPLEY-PROSPECT HTS SUBSTATION 100 kV L
Contingency: STAPLEY-ARLINGTON HTS 100 kV
Loading: 125.0% (Rating: 165 MVA)
DFAX: 29.7%
Pre-project loading: 92.2%
5. Facility: GOOSE HAVEN-ROSEMORE B 100 kV Line 1
Contingency: GOOSE HAVEN-ROSEMORE B 100 kV
Loading: 122.9% (Rating: 248 MVA)
DFAX: 35.3%
Pre-project loading: 96.9%

6. Facility: CUMBERLAND WI-ARLINGTON HTS 100 kV
 Contingency: STAPLEY-PROSPECT HTS 100 kV line
 Loading: 115.8% (Rating: 112 MVA)
 DFAX: 23.6%
 Pre-project loading: 77.3%
7. Facility: ARLINGTON HTS-STAPLEY 100 kV L
 Contingency: STAPLEY-PROSPECT HEIGHTS 100 kV
 Loading: 101.5% (Rating: 170 MVA)
 DFAX: 23.6%
 Pre-project loading: 76.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
STAPLEY-ROSEMORE B 100 kV Ckt 1	Rebuild line with higher-capacity conductor	\$2,606,000	57% (\$1,489,000)
STAPLEY-ROSEMORE B 100 kV Cir. 2	Rebuild line with higher-capacity conductor	\$1,818,000	57% (\$1,038,000)
GOOSE HAVEN-ROSEMORE B 100 kV Ckt 2	Rebuild line with higher-capacity conductor	\$1,774,000	57% (\$1,013,000)
PROSPECT HTS-STAPLEY 100 kV	Rebuild line with higher-capacity conductor	\$1,663,000	55% (\$910,000)
GOOSE HAVEN-ROSEMORE B 100 kV Line 1	Rebuild line with higher-capacity conductor	\$1,159,000	57% (\$662,000)
CUMBERLAND WI-ARLINGTON HTS 100 kV	Rebuild line with higher-capacity conductor	\$583,000	68% (\$398,000)
ARLINGTON HTS-STAPLEY 100 kV L	Reconductor line	\$259,000	68% (\$177,000)
	Total allocated to project		\$5,687,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.